

Virtual reality: the gateway for non-professionals to solve complex design problems

Dorottya Esztergalos

Leopoly, Hungary

dorottya.esztergalos@leopoly.com

Marcell Pal

Leopoly, Hungary

marcell.pal@leopoly.com

Daniel Andrassy

Leopoly, Hungary

daniel.andrassy@leopoly.com

Zoltan Karpati

Leopoly, Hungary

zoltan.karpati@leopoly.com

ABSTRACT

Leopoly is working on a 3D design software for VR called Shapelab with the aim to provide tools for professional and hobby 3D artists. During development, we have been approached by users coming from various professional backgrounds other than artistic 3D design, which made us want to explore the possibilities virtual reality has to offer for these use cases. We argue that VR could open doors for people with no background in 3D modeling to solve complex design problems needed to digitize their traditional workflows.

ACM Reference Format:

Dorottya Esztergalos, Daniel Andrassy, Marcell Pal, and Zoltan Karpati. 2023. Virtual reality: the gateway for non-professionals to solve complex design problems. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educator's Forum (SIGGRAPH '23 Educator's Forum)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3587424.3595582>

1 INTRODUCTION

The demand for 3D digital assets is on the rise and more businesses are looking for opportunities to digitize their production workflow with the help of 3D printing and scanning technologies. However, creating 3D assets requires a professional level of understanding of 3D modeling tools as well as years of practice. This is where virtual reality technology comes into the picture. The AR/VR/MR market size is expected to rise to over 250 billion U.S dollars by 2028 [Statista, 2023], and while virtual reality technology is mostly associated with the video game industry, its application can already be noticed in other fields where the need for 3D assets is growing, such as art and design, healthcare or education. According to previous research, modeling objects in VR is efficient and effective [Van Goethem et al., 2020] promotes creativity and a greater feature creation rate [Freeman et al., 2018], and offers better spatial perception and visualization [Vlah et al., 2021]. Our team reviewed user feedback for our VR modeling application and found that it resonated with these findings.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SIGGRAPH '23 Educator's Forum, August 06–10, 2023, Los Angeles, CA, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0146-7/23/08.

<https://doi.org/10.1145/3587424.3595582>

2 OUR HYPOTHESIS

We strongly believe that virtual reality can be integrated into the design workflow to increase productivity and give users a more intuitive and enjoyable experience than traditional flat-screen solutions. After gathering feedback and testimonials from our users, we argue that VR could open doors for people with no background in 3D modeling to solve complex design problems needed to digitize, and therefore improve their traditional workflows and processes.

3 STORIES OF MOVING FROM PHYSICAL TO DIGITAL

Photogrammetry, 3D scanning, and 3D printing technologies have evolved tremendously in the past years, paving the way for integrating digital into traditional workflows in fields such as product design, art, film and entertainment industry, forensics, and medical research. The latter is a great example of how 3D digital technologies have contributed to the evolution of the field. We talked to a clinical director who specialized in Orthotics and Prosthetics to gain some insight into how their workflow evolved from a traditional to a digital approach and all the challenges that came with it.

Traditionally the cast for the orthosis or prosthesis was created from a mold that had to be adjusted physically to the body shape of the patient. This process became faster and more flexible thanks to 3D scanning, which allowed the cast to be digitally created and modified, with the following process: after 3D scanning the body part, the scan is processed in order to create a clean mesh that then can be digitally modified in ways to allow for a comfortable, effective and custom fit device that the patient can stand on or use functionally. The product is then designed to suit the adjusted 3D-scanned body part. It is 3D printed and further processed by applying additional layers of various materials. The final product is completed with parts such as mounts or joints. The digital process is more accurate and more efficient, and most importantly, it enables consistent repeatability, as orthoses made with 3D printing are infinitely reproducible. Digitally created products are also easier to customize and the mesh can be optimized in terms of structure, aesthetic, or material. Moreover, the digital workflow contributes to less waste being created by eliminating subtractive milling from the process. For these reasons, digital methods will likely replace traditional ones in a few years, which will present challenges for those clinicians who have not found an adequate way to translate their skillset into the digital age.

During the past year, we have also learned about other fields where digital 3D design was a welcomed, but challenging development: indie game developers, sculptors, and other artists, many hobby designers, and even a glass worker have reached out to us hoping to find a solution for adopting 3D design. We have gathered a few areas that present a challenge for these users using traditional 3D tools.

4 ADOPTION CHALLENGES

4.1 Learning 3D modeling

For beginners in 3D design, complex 3D software takes time to learn because of a huge variety of commands and the difficulty of understanding perspective while having to navigate in a 2D environment with their mouse and keyboard. For many artists, physical workers, and specialists experimenting with 3D, this is a significant barrier to entry that leads to abandoning these ambitions. Most clinicians in the Orthotic and Prosthetic field come from a craftsman-style background rather than traditional engineering, so there has been a general hesitancy from a majority of the field to switch over from traditional plaster. Many clinicians understand that eventually CAD modifying could speed up fabrication, but can't put in the time investment to develop their skills.

4.2 Editing 3D scan data

Specifically for 3D scanning and photogrammetry we see the challenges in two areas of the data editing workflows: cleaning the heavy scan data and modifying the mesh:

- Editing 3D scan data is cumbersome work, especially for people who have less routine with traditional desktop 3D software. Cleaning the heavy scan data from floating geometry and patching holes requires a thorough inspection and takes a long time for users who are not used to navigating in 3D on a flat screen with a keyboard and a mouse.
- Modifying the mesh to create the final product requires artistic abilities, precision, and a good grasp of perspective if done on a desktop screen. This is a long learning process that also needs routine, which is time-consuming.

5 VIRTUAL REALITY AS A FACILITATOR FOR ADOPTION

We asked our users to share their thoughts on how VR design tools and Shapelab were helping them in acquiring digital design skills in their work and hobby projects. The following areas were mentioned:

- VR interactions make 3D navigation intuitive. Virtual reality provides spatial vision and an understanding of perspective which is a huge benefit compared to traditional software. The ability to engage with the work at life size or larger gives a truer sense of the subject and being able to work in broad gestures is much freer than on a flat screen, according to our users coming from various fields such as traditional sculpting and healthcare.
- VR interactions also make 3D navigation quick. Artists use VR tools to quickly block out basic shapes much faster due

to freedom of movement and size. VR could simplify a complicated scan data editing process as heavy scan data can be easier and more intuitive to inspect in VR, which also makes the cleaning process faster.

- VR tools are handy for people with craftsman-style backgrounds. Those who already possess hand skills can transition more smoothly to a digital era as VR provides them with a more life-like sculpting experience.
- Virtual reality can also offer a more enjoyable experience, not only for beginners but for professionals as well. Users such as XR artists and indie game developers might try VR instead of traditional 3D software, as they feel it is a more entertaining and intuitive way of creating. They can quickly become proficient which gives them a sense of accomplishment.

6 THE UNIQUENESS OF SHAPELAB

Shapelab is unique among VR creative tools in having a polygon mesh-based engine instead of a voxel-based one, which is usually more suited for designing characters and digital props. We believe that VR technology and Shapelab's robust engine and feature set can provide users of different backgrounds with tools to effectively create 3D assets.

- For professional 3D artists, features such as dynamic topology mode and subdivision levels offer control over polygons like no other VR creative application and allow users to create very detailed and realistic 3D objects, as well as low-poly models.
- For newcomers to 3D design, we maintain an intuitive interface so that the user can focus on creating.
- Shapelab's robust engine allows for even heavy scanned data to be processed. Users looking to digitize their workflow can import and edit 3D scans and export 3D printable models.

7 THE EVOLUTION OF VIRTUAL REALITY IN 3D DESIGN

VR design tools are currently not advanced and powerful enough to completely replace well-known and robust flat-screen solutions for professionals working in 3D modeling. However, the VR market is quickly evolving and applications such as Shapelab are implementing more features to expand the VR workflow.

We are hoping that virtual reality could lower the barrier of entry for a wider audience in the future – both aspiring creatives, as well as specialists coming from non-design related fields.

REFERENCES

- Seth M. Freeman, Landon B. Wright and John L. Salmon. 2018. Exploration and evaluation of CAD modeling in virtual reality. *Computer-Aided Design and Applications*. 2018, VOL. 15, NO. 6, 892–904, <https://doi.org/10.1080/16864360.2018.1462570>
- Statista. XR/AR/VR/MR market size 2021-2028, Retrieved from: www.statista.com, February 15, 2023 from: <https://www.statista.com/statistics/591181/global-augmented-virtual-reality-market-size/>
- Daria Vlah, Vanja Čok and Uroš Urbas. 2021. VR as a 3D Modelling Tool in Engineering Design Applications. *Applied Sciences* 11, no. 16: 7570. <https://doi.org/10.3390/app11167570>
- Sander Van Goethem, Regan Watts, Arno Dethoor, Rik Van Boxem, Kaz van Zegveld, Jouke Verlinden and Stijn Verwulgen. 2020. The Use of Immersive Technologies for Concept Design. *Advances in Usability, User Experience, Wearable and Assistive Technology* (pp.698-704), https://doi.org/10.1007/978-3-030-51828-8_92